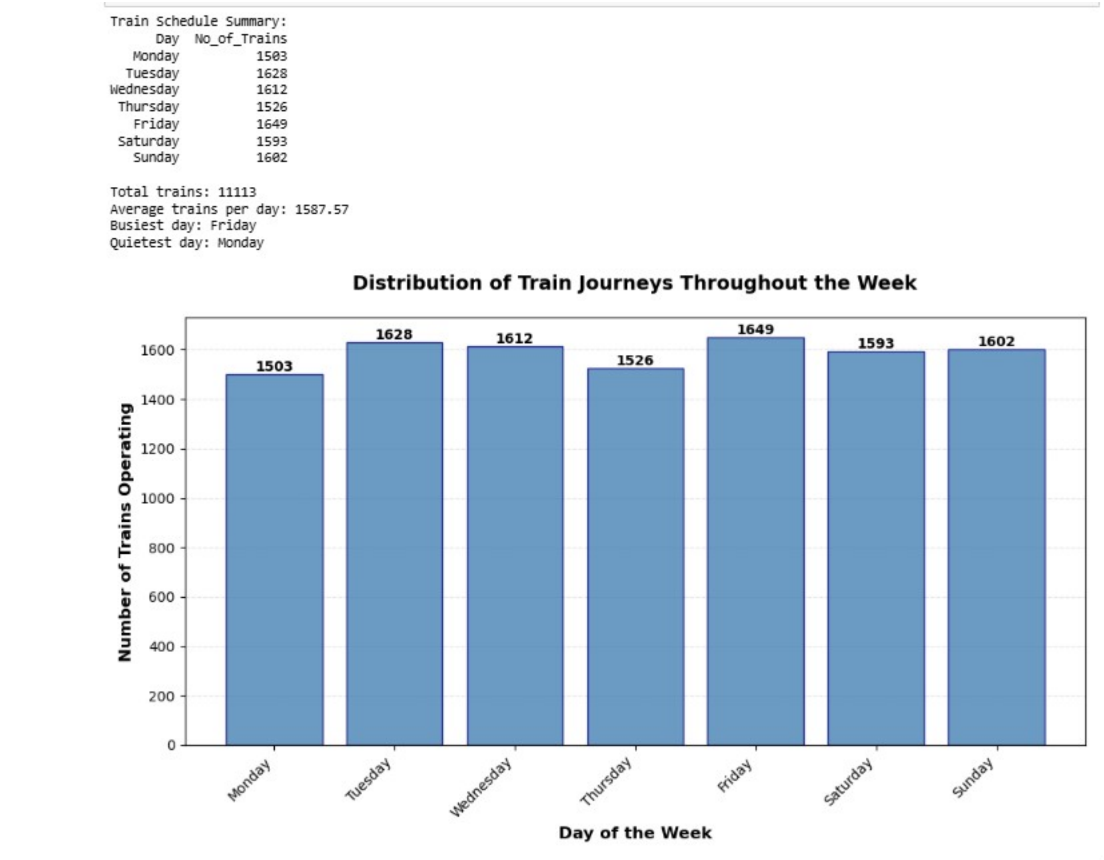
Final Internship Report – Railway Data Analysis

Key Observations :-

- **Hub Concentration:** A few major stations handle most train departures, showing hub-and-spoke connectivity.
- **Busiest Days:** Weekdays have higher train frequencies compared to weekends.
- **Top Source:** CST-Mumbai is a major hub with maximum outbound connections.
- **Service Patterns:** Express and Superfast trains dominate; Passenger and Mail trains run more evenly across days.
- **Top Routes:** Specific station pairs (from top 10 OD heatmap) show highest traffic density.

Visual Evidence.



Bar Chart: Trains per weekday (shows busiest and quietest day).

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Day Type Summary:
  Day_Type  No_of_Trains
0  Weekday          7918
1  Weekend          3195

Total Trains: 11113
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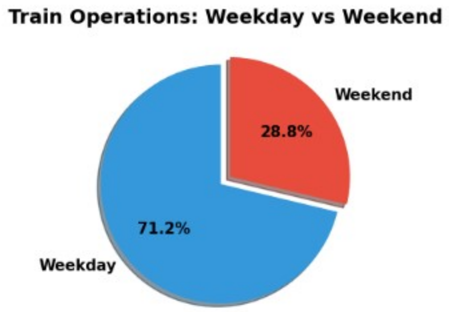
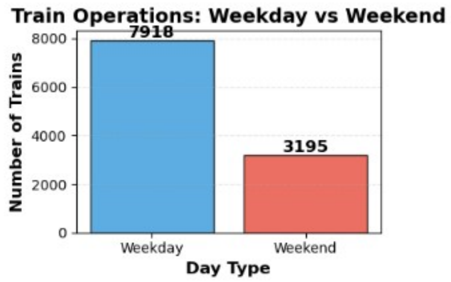
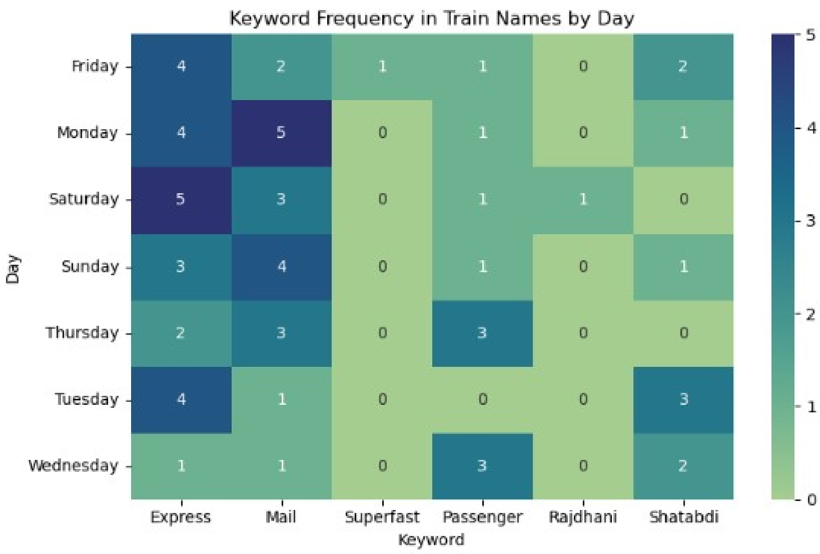
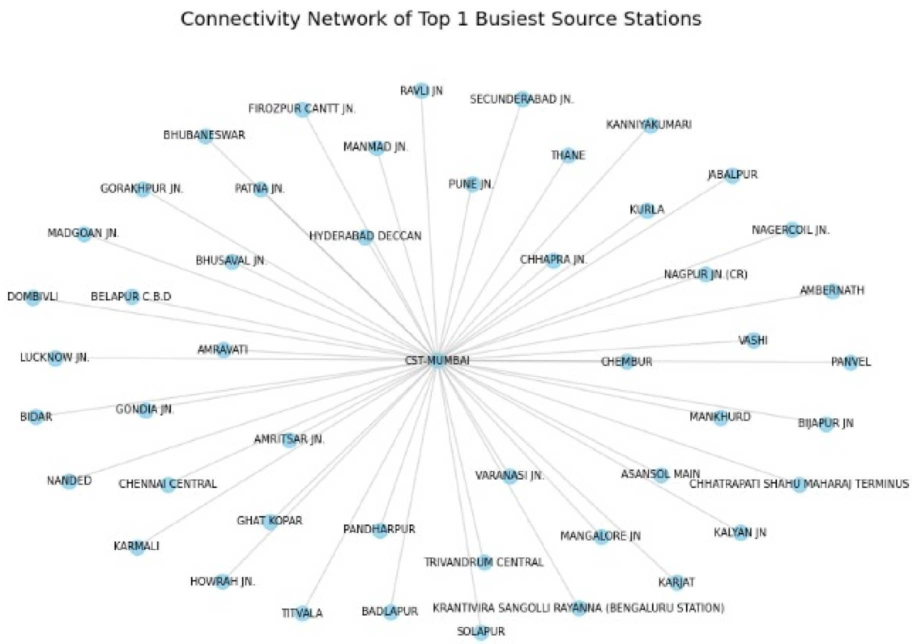




Pie Chart: Day-type distribution (Weekday, Weekend, All-days).

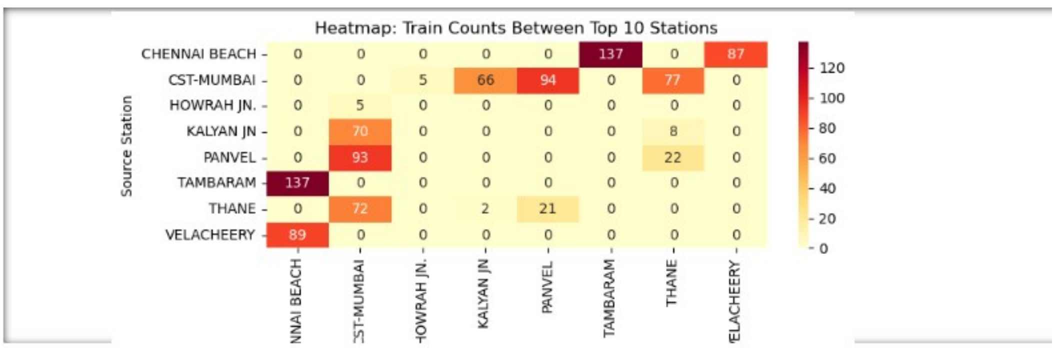


Heatmap: Train type frequency by day (Express, Mail, Superfast, etc.).



Network Graph: Connectivity from top source station.

Bar Chart: Top 10 source stations.



Heatmap: Train count between top 10 source–destination pairs.

Recommendations

Optimize train schedules on high-traffic weekdays.

Improve infrastructure at top hub stations.

Add capacity on busiest routes.

Use service-type insights to plan weekday vs weekend operations.

Extend analysis with ridership and punctuality data for better planning.

Summary

The analysis highlights strong operational hubs, weekday peaks, and distinct service patterns. Visual trends can guide scheduling, resource allocation, and network optimization for improved efficiency.